

NEW PRODUCTS

MAKE IN INDIA

COMPONENTS

PIC and AVR microcontrollers

Ideal for applications that use closed-loop control, PIC18 Q10 and ATtiny1607 microcontrollers (MCUs) have core independent peripherals (CIPs) that independently manage tasks and reduce the amount of processing required from the CPU. System designers can save time and simplify design efforts with hardware-based CIPs, which significantly reduce the amount of software required to write and validate. Both families have features for functional safety and operate up to 5V.

Microchip Technology
www.microchip.com

Power modules

Infineon Technologies offers IM512 and IM513 series of CIPOS Mini family.

These intelligent power modules integrate CoolMOS MOSFETs for excellent electronic performance. These help in optimising power efficiency in applications such as compressors, pumps or fans for home appliances.

The modules are especially suitable when a system operates under light-load conditions. These offer low on-resistance of 310mΩ and a current rating of 10A at 25°C, with a breakdown voltage of 600V.

Infineon Technologies India Pvt Ltd
www.infineon.com

Medical DC-to-DC converter

Traco Power has released THM 20 family of medical 20-watt DC-to-DC converters with a wide 2:1 input range. The converters provide high isolation of 5000V ACrms (250V ACrms working

voltage) with reinforced insulation and low leakage less than 2.5uA. Features include extended -40°C to +80°C operating temperature range (full power to +55°C), efficiencies up to 90 per cent, ±10 per cent voltage trim range, EMC compliance to IEC60601-1 4th edition and EN55032 class A, operating altitude up to 5000m, under-voltage lock-out, overload protection and short-circuit protection.

Traco Power
www.tracopower.com

Buck converters

Automotive designers can significantly lower electromagnetic interference and increase efficiency with pin-compatible MAX20073 and MAX20074 buck converters from Maxim Integrated Products.

Key features are:

Lowest EMI. Unique spread-spectrum and sync-in features mitigate system EMI radiation

Higher efficiency. Exhibits superior efficiency across the entire operating range for best-in-class efficiency

Small solution size. Small 3mm x 3mm 10-pin TDFN-EP package reduces overall system footprint

Output voltage accuracy. Output voltage can be pre-set at the factory to allow ±1.5 per cent output voltage accuracy without 0.1 per cent external resistors

Maxim Integrated Products
www.maximintegrated.com

Insulated-gate bipolar transistors

ROHM has announced two new types of 650V insulated-gate bipolar transistors that combine low conduction loss with high-speed switching characteristics. This makes these ideal for power conversion in general-purpose inverters

and converters for consumer appliances, such as air-conditioners and induction heaters, as well as industrial equipment, including power conditioners, welding machines and UPSes.

ROHM Semiconductor India Pvt Ltd
www.rohm.com



Arm Cortex M4 MCUs

Nuvoton has launched its high-performance NuMicro M480 Series Arm Cortex M4 MCUs with secure boot function and hardware cryptography. M480 series consists of six sub-series targeting different applications. It supports up to 512kB flash and 160kB SRAM. Operation frequency is up to 192MHz with 175μA/MHz dynamic power consumption, and standby current can be as low as 1μA. Secure boot functionality provides a constant digital signature of system software for identification during boot up to protect the integrity of flash data from malicious attack.

Nuvoton Technology Corp.
www.nuvoton.com

Gas sensor module

Toshniwal Sensing Devices introduces CO₂ gas sensor module CDM-7160 from Figaro, Japan. This sensor can be used in indoor air quality-control systems, including variable-air volume systems



and demand-controlled ventilation systems. It uses a compact NDIR CO₂ sensor, providing high accuracy at low power. Two detector elements inside the module make absolute measurement possible. The module comes with both UART and I2C digital interfaces.

Toshniwal Sensing Devices Pvt Ltd
www.tsdppl.com